Predictive Machine Learning System for CyberArk Product Sales

DISSERTATION

Submitted in partial fulfillment of the requirements of the

Degree : MTech in Artificial Intelligence and Machine Learning

By

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2023AA05203

Under the supervision of

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DSECLZG628T / **AIMLCZG628T DISSERTATION**

Dissertation Title: Predictive Machine Learning System for CyberArk Product Sales

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Name of Student: Saya Venkata Thrivedi

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Courses Relevant for the Project & Corresponding Semester:
1. Machine Learning
2. Artificial and Comp Intelligence
3. MLOPS
4. Deep Neural Networks

Abstract

In the competitive world of enterprise cybersecurity sales, smart, data-driven ways are needed to find high-potential customers and improve sales efforts. Regular sales forecasting often struggles to use the many different company and customer behaviors that lead to a purchase. This project tackles this key challenge by creating a new machine learning system for predictions. This system is designed to accurately find enterprise customers who are very likely to buy cybersecurity solutions, helping sales teams work smarter and potentially boost sales.

The system will use a varied dataset with important business details like company size, yearly income, industry, security readiness, past security breaches, number of privileged users, identity tools, cloud platform use, decision-maker roles, engagement scores, and lead source information. The main goal is to build a strong system that accurately predicts if a customer will buy or not.

The project will start by building basic models (like Logistic Regression and Decision Trees) to see how well they perform and understand which features matter most. Then, these will be compared with more advanced models like Random Forest, XGBoost, and CatBoost. These are good at finding complex patterns and making more reliable predictions. Lastly, neural networks will be explored to see if they can handle the very complex data often found in enterprise sales.

Models will be thoroughly checked using standard measures like precision, recall, F1-score, and ROC-AUC. Importantly, techniques like SHAP will be used to explain *why* the model makes certain predictions, giving sales and marketing teams clear, useful information about what makes customers likely to buy. The best model will be picked based on how well it predicts, how well it works on new data, and how easy its predictions are to understand.

The system will be put into action with a simple, interactive user interface for quick predictions. The entire process, from preparing data to training, evaluating, and deploying the model, will be packaged using Docker and set up with CI/CD tools for ongoing updates and delivery. Over sixteen weeks, this project will deliver a practical, easy-to-understand, and ready-to-use machine learning solution. It will combine careful model testing with good MLOps practices, showing how advanced prediction models can be used in real-world cybersecurity sales.

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DSECLZG628T / **AIMLCZG628T DISSERTATION**
Dissertation Outline (Abstract)

BITS ID No. 2023AA05203 **Name of Student:** Saya, Venkata Thrivedi

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Topic of Dissertation: Predictive Machine Learning System for CyberArk Product Sales

(Signature of Student) (Signature of Supervisor)

Date: 21-May-2025 Date:----------

Broad Area of Work

The project falls under the broad domain of Artificial Intelligence (AI) and more specifically focuses on the subfields of Applied Machine Learning (ML), Predictive Modeling, and Enterprise Decision Intelligence. In today's data-driven ecosystem, organizations increasingly rely on machine learning systems to make informed strategic decisions, including the prediction of customer intent and purchase behavior. This project contributes to this broader vision by applying supervised learning techniques to predict the likelihood of cybersecurity product purchases by enterprises based on their digital and organizational characteristics.

The work intersects multiple thematic areas including:

- *Business Intelligence and Predictive Analytics:* Leveraging historical and real-time enterprise-level data to forecast potential purchasing intent.
- *Model Evaluation and Selection*: Implementing and benchmarking classical, ensemble-based, and neural network models for binary classification tasks.
- *MLOps and Model Lifecycle Management*: Integrating DevOps practices into the ML pipeline to ensure reproducibility, scalability, and maintainability of the deployed solution.
- *Human-Centered AI Interfaces:* Delivering model predictions through a simple, user-friendly web interface for decision support.

This interdisciplinary project blends concepts from computer science, statistics, and enterprise sales operations to build a robust and explainable ML-driven recommendation system that can assist cybersecurity product teams in identifying high-value customer prospects efficiently.

Objectives

The key objectives of this project are as follows:

- To explore and preprocess enterprise sales data capturing organizational attributes, security readiness, and engagement metrics.
- To design and implement a suite of predictive models (baseline, advanced, and neural network-based) for forecasting the likelihood of cybersecurity product purchases.
- To systematically compare the performance of models using appropriate classification metrics (Precision, Recall, ROC-AUC, F1 Score).
- To select and fine-tune the best-performing model for deployment.
- To build a simple, interactive user interface for prediction demonstration.
- To containerize the solution using Docker and implement CI/CD for automated retraining and deployment.

Scope of Work

The scope of this dissertation includes the end-to-end development of a machine learning pipeline to solve a real-world problem in enterprise product recommendation and intent prediction. The project involves:

- Preprocessing a diverse, real-world styled dataset containing enterprise, security, and engagement attributes.
- Building and evaluating predictive models such as Logistic Regression, Random Forest, Gradient Boosting, and Feedforward Neural Networks.
- Designing a UI for interacting with the final predictive model.
- Enabling model reproducibility and deployment using containerization (Docker) and optionally integrating CI/CD pipelines for model retraining and monitoring.

Detailed Plan of Work (Sample) (for 16 weeks)

Serial Number of Task/Phases	**Tasks or subtasks to be done** (be precise and specific)	Start Date-End Date	Planned duration in weeks	Specific Deliverable in terms of the project
1	Requirement analysis and data understanding	Week 1	1	Requirement doc, data schema report
2	Data cleaning, exploratory data analysis (EDA), data validation	Week 2-3	2	Dataset Cleaning and EDA Report
3	Feature engineering and data preprocessing (scaling, encoding, etc.)	Week 4-5	2	Data Preprocessing
4	Implementation of baseline models (Logistic Regression, Decision Tree)	Week -6	1	Baseline model results
5	Implementation of advanced models (Random Forest, XGBoost, etc.)	Week 7-8	2	Advanced model results
6	Implementation of neural network models (MLP, tuning, etc.)	Week 9	1	Neural model results
7	Model comparison and selection using metrics and cross-validation	Week 10- 11	1	Model Evaluation report
8	Model Export & UI development (simple web interface for predictions)	Week 12	1	Working UI integrated with best model
9	Dockerization of the application	Week 13	1	Docker container for complete app
10	CI/CD pipeline design and automation (GitHub Actions / Jenkins)	Week 14	1	CI/CD pipeline design and automation (GitHub Actions / Jenkins)
11	Final testing, documentation, and presentation preparation	Week 15 - 16	2	Final report and presentation

Literature References

The following are referred journals from the preliminary literature review.

[1] Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 2019.

[2] Choudhury, T., *Practical MLOps*, O'Reilly Media, 2021.

[3] Ribeiro, M. T., Singh, S., & Guestrin, C., *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*, KDD, 2016.

Supervisor's Rating of the Technical Quality of this Dissertation Outline

EXCELLENT / GOOD / FAIR/ POOR (Please specify): _____

Supervisor's suggestions and remarks about the outline (if applicable).

Date 21-May-2025

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(Signature of Supervisor)

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